

ORIGINAL RESEARCH

Delay aware resource allocation in ORAN through network optimization

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Abstract

A multi variable resource allocation problem is investigated in network environments, specifically focusing on the consideration of quality of service in open radio access network. The main objective is to minimise the combined latency of various servers while complying with network limitations. The delay of each server is represented by a non-linear function that has exponentially based. This characteristic inherently brings non-convexity into the objective function. In contrast, the constraints comprise various linear combinations of network variables, including resource block allocations, power consumption, and number of virtual machines. The purpose of these constraints is to guarantee that the allocation of resources adheres to practical limitations and upholds fairness among servers. Nevertheless, the inclusion of a non-convex objective function significantly adds complexity to the optimisation problem and non-convex behaviour, requiring specialised algorithms and techniques to identify solutions. Subsequently, the Lagrange multiplier method has been used to solve this problem mathematically. Numerically, three algorithms have been utilised and compared to solve the problem, these are active-set, interior point and sequential quadratic programming. Note that the total delay as an objective function is based on the total power consumption of the servers. Previous to optimising the total delay, a delay model is proposed and compared with two research works that are based on experimental and real time data. The proposed model shows data matching with the other works and permits for more adaptation/integration with any other works that uses different servers' characteristics and network parameters.

KEYWORDS

delays, mobile communication, optimisation

1 | INTRODUCTION

The field of wireless communication has experienced a significant revolution with the evolution of open radio access networks (ORAN), which signifies a fundamental shift in how cellular networks are designed, deployed, and managed. ORAN, distinguished by its disaggregated and open design, is fundamentally transforming the conventional closed and proprietary structures that have historically delineated the cellular environment [1]. This research paper examines the emergence

of ORAN, providing an in-depth analysis of its underlying concepts, architectural principles, and the extensive implications it has for the advancement of wireless communication infrastructures.

The primary constituents of ORAN encompass radio units (RUs), distributed units (DUs), centralised units (CUs), open interfaces, management and orchestration, and virtualisation as shown in Figure 1. The ORAN architecture is primarily focused on achieving several key objectives. These objectives encompass the reduction of network deployment costs, enhancement of

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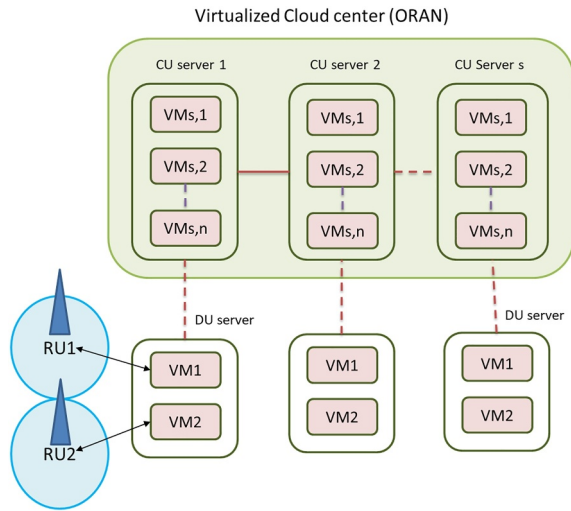


FIGURE 1 Open radio access network architecture.

flexibility, promotion of innovation through open interfaces, facilitation of multi-vendor interoperability, and acceleration of the introduction of new services inside mobile networks, particularly in the context of the 5G and beyond.

ORAN, at its fundamental essence, encompasses the fundamental tenets of openness and standardisation, hence facilitating the emergence of a telecommunications market characterised by enhanced diversity and competitiveness. The introduction of ORAN enables a higher degree of agility in traditional cellular networks by separating hardware and software components. The process of disaggregating network elements, including RUs, baseband or CU units, and network functions, enables the utilisation of a mix-and-match strategy. This technique facilitates the integration from many vendors by utilising standardised protocols.

The emergence of 5G technology and the potential for future developments, such as 6G, have generated an increasing need for networks that possess not only superior performance but also cost-efficiency, and versatility to accommodate various applications [2]. The ORAN concept presents a unique architectural framework that is in line with the aforementioned requirements. It facilitates a dynamic ecosystem consisting of vendors, developers, and operators. It is crucial for operators and industry stakeholders to possess a comprehensive comprehension of the complexities and ramifications associated with ORAN. Resource allocation is a critical factor in optimising network performance and maximising the optimal use of available resources within the dynamic and ever-changing realm of wireless communication systems. Resource allocation challenges play a crucial role in improving system efficiency, enhancing quality of service (QoS), and optimising overall network performance. These problems encompass a range of tasks, including bandwidth allocation, power optimisation, delay and latency, and network parameter management. Nevertheless, the intricate and ever-changing characteristics of wireless networks give rise to difficulties that beyond conventional linear optimisation frameworks. This study explores

the field of non-convex optimisation within the domain of minimising the delay of the evolutionary architecture, represented by ORAN, with a specific emphasis on the complexity that develop while addressing these sophisticated problems [3].

The primary goal in a wireless communication system is to effectively distribute resources, including spectrum, power, and computational capability, among different users and services, while ensuring that particular quality-of-service criteria are met [4]. The allocation decisions pertaining to resources have a significant impact on the basic criteria that determine the network performance, such as data rates [5], latency [6], and energy efficiency [7]. The conventional method for addressing resource allocation issues has frequently relied on convex optimisation, capitalising on its inherent simplicity and effectiveness. Convex optimisation is distinguished by its properties of possessing convex goal and constraint functions, which facilitate the utilisation of existing mathematical methodologies, such as linear programming, to effectively identify global optima [8].

The presence of sophisticated system dynamics typically gives rise to complexities in the form of non-linear, non-convex objective functions and constraints. The problem discussed in this paper exemplifies a scenario in which the objective function is non-convex as a result of the inclusion of exponential elements in the total delay function. This study aims to analyse and investigate the challenges and subtleties associated with minimising the total delay of the ORAN network subject to different constraint including: resource blocks (RB), initial and maximum power consumptions of the VMs, initial and maximum power of the server and the number of VMs in each server. By employing rigorous analysis and empirical studies, the work intends to provide significant insights that might inform the design of forthcoming wireless communication systems. Prior to the complex realm of non-convex minimisation, a model to measure the power has been proposed mathematically based on the power consumption of the network. In which, the fine network factor such as number of RB and bits; can impact the outcome of the delay model. The proposed model also allows for enlarged number of modification to cope with the type of the network, application, service and load status.

1.1 | Related works

Although ORAN architecture has promised minimised delay as part of the 6G requirements [9]. Its architecture can add extra delay due to the invention of middle boxes, represented by DUs. In this context, there are many reduction phases that can be achieved. One example is using advanced and evolved resources optimisation techniques as proposed in ref. [10] using Fog radio access network. It should be noted that Fog networks share a similar architecture with ORAN. However, it is important to highlight that Fog networks do not possess an open framework that allows for participation from all vendors, operators, and manufacturers. Fog networks employ proximate nodes to fulfil the essential duties of users in real-time situations [11]. Another solution is using ultra fast electronic

devices, such as in ref. [12], where ADC converter and radio frequency units are enhanced, by which, the processing of data became faster.

Note that the latency factor is crucial in the coming generation and has been manifested in several ORAN-related researches, such as refs. [9, 13]. Practically, Fog networks is the legacy version of ORAN architecture. By advancing the networks with open software and hardware, ORAN has evolved. The latency in Fog network has been studied in many researches. In ref. [14], the radio and cache resource management are discussed. Network slicing is a cost-efficient method for operators to provide as-a-service networks. This work focused on network slicing including a unified orchestration and a slice instance layer. The authors of ref. [15] proposed fog radio access network architecture to examine the need for extreme latency critical services in Internet of things (IoT) field. By employing a distributed computing approach, fog computing has the capability to fulfil the specific demands of certain applications. The work examines the necessity of a service framework that can effectively manage performance, computational cost, and communication cost. Subsequently, in ref. [15], fog radio access network architecture is introduced to meet an ultra low-latency communications services in IoT applications. By distributing computing-heavy tasks to several nodes, fog can meet the requirements of certain applications. The article discusses the need for a service framework to balance communication cost, computing cost and performance, and illustrates mobile services as an exemplary scenario. This work lacks mathematical modelling for the proposed problem.

In order to achieve ultra-low latency and solve the problem of heavy burden on fronthaul capacity, the authors of ref. [16] introduced a loosely coupled architecture in Fog network. This way of architecture enables multiple Fog nodes' cooperative distributed computing and content sharing to meet the minimum latency requirement. The proposed scheme employs a greedy algorithm and the WMMSE technique in order to maximise the transmission rate. This study provides a holistic overview for delay assessment techniques employed within optimisation problems. Nevertheless, the model offered is simplified and neglects to capture the exponential behaviour that characterises certain network applications.

Recently, the focal point of concern is directed towards the subject of ORAN in the forthcoming network generations. It is crucial to gain a comprehensive understanding of the latency behaviour of the subject matter by thorough study and evaluations. In addition, the provision of a novel latency model is of utmost importance in order to establish a comprehensive framework that may be utilised by researchers who may not prioritise the measurement of delay, yet nevertheless necessitate knowledge of the delay for application in other contexts. Intelligent traffic steering scheme in ORAN is discussed in ref. [17] with discussions on intelligent traffic prediction and radio resource management. The problem is formulated by the long-term and short-term subproblems, and as for the solved long-term subproblem, the specific model used is the long-short-term memory model.

However, this work was not aimed directly to evaluate the delay. Since the limitations of 5G and 6G services come from the legacy functionality, the authors of ref. [18] proposed ORAN architecture as a solution. In which, they focused on machine learning model-based network functionality to optimise user handover and service resource orchestration.

The paper under study [19] focuses on a service-aware baseband resource allocation and virtual network function activation for the ORAN system, which should be performed for network slicing. It partly solves the problem of a lack of fronthaul capacity and end-to-end delay requirements. The problem is characterised as mixed-integer programming problem, but an iterative algorithm is proposed for near-optimal solutions. The present study exhibits an incomplete depiction of the ORAN delay as a result of its emphasis on channel capacity constraints. In the same manner as provided in ref. [20], this paper applies a joint optimisation approach to ORAN architecture to improve energy efficiency and to offer an appropriate delay bound to the users. Optimisation is performed using the software that solves the mixed-integer linear problems. The optimisation problem involves virtualised processing pools with limited capacity, affecting network performance and energy consumption.

To follow, in ref. [21], it was assumed that the traffic load in mobile network, specifically referred to service function requests is directly or exponentially proportional to the delay of the VMs; in case of single path or multi-path models, respectively. In ref. [22], an analytical model is proposed and evaluated, employing single share buffer for handover process. The results indicate that the delay experienced by networks with a larger load and packet-in messages probability exhibits a linear growth. In ref. [11], the fog network is regarded as having distinct characteristics in terms of delay, with linear behaviour observed under light traffic conditions and exponential behaviour observed under heavy traffic conditions. In the context of the ORAN network, it is anticipated that the delay will exhibit similar characteristics of Fog, cloud and traditional RAN, when accounting for various network metrics and traffic types. In ref. [23], the investigation of LTE network latency in a virtualised environment was conducted, focusing on the relationship between latency and payload size. It was observed that the latency exhibits an exponential increase as the payload size increased, eventually reaching a maximum threshold. The utilisation of the server has exponential relationship with the delay. Moreover, it has been noted in the literature [24, 25] that the latency of the servers is influenced by the load in either a linear or exponential manner.

2 | DELAY ASSUMPTIONS

In this work, we proposed a simplified power model, yet descriptive, to model the delay behaviour. Experimental latency models are still valuable tools for studying network behaviour, but they have several disadvantages. They are resource-intensive, requiring significant computational resources, making them expensive and time-consuming to run. Realism in

these models can be challenging because of the dynamic behaviour of real-world network conditions. Scalability is also a challenge, as models may not scale well to large networks or scenarios with numerous network elements. Results from these models may not always generalise well to different network configurations or environments. Creating accurate models often requires access to extensive network measurement data, which may not always be available or up-to-date. Complex and functions' based models may not always provide accurate predictions about future network performance, especially when unforeseen events or changes occur. Validating these models against real-world data can be challenging, as it may require access to specific network configurations and conditions. In conclusion, based on these facts: there are some facts to be considered to model ORAN latency model:

1. A direct or exponential correlation exists between the workload and the latency of the servers. This issue can be represented using a growth model; however, it is vital to notice that delay values vary depending on the characteristics of different servers. Furthermore, the translation of workload into relevant parameters within the LTE paradigm, such as the number of received packets, RB, or amount of bits, is not adequately addressed. Thirdly, the magnitude of the workload has varying effects on the delay, indicating that a large workload might exert a significant and immediate impact on network delay, hence introducing unrealistic modelling considerations.
2. The servers' delay exhibits exponential characteristics in relation to their power utilisation. Maximising server utilisation leads to an increased volume of received workload. This implies that certain packets are needed to wait for a period of time in order to allow for the processing of other packets. This principle holds true regardless of whether the server is virtualised or not. The modelling of this behaviour is not possible in a straightforward manner due to the generic nature of its utilisation.
3. There is direct or exponential relationship between the power consumption and the workload. In addition, there is a limit to the servers' consumption based on each server's characteristics. In this matter, the workload can be translated to more understood parameter and such behaviour can be modelled.
4. The number of VMs placed within a server has a direct linear impact on both power consumption and delay.

3 | DELAY MODELLING

The proposed methodology involves establishing a mathematical representation of the delay in relation to power consumption. Subsequently, the power consumption is characterised in terms of the VMs, then the VMs are modelled as a function of workload and integrated back into the delay model.

3.1 | CU delay

Case 1. The delay is a function of power consumption.

Since there is no upper limit for the delay increment, the growth model is valid candidate, where the change in the delay τ is proportional to the power consumption PC as follows:

$$\frac{\partial \tau}{\partial PC} = \alpha \tau(PC) \quad (1)$$

However, this linear power model can lead to a non-linear model that has exponential behaviour as a result, as explained in the following manner:

$$\tau(PC) = \tau_0 e^{\alpha PC} \quad (2)$$

where α is the delay increasing constant factor.

Case 2. The relationship between power consumption and the number of VMs is exponential, with a maximum limit imposed on the power. In this modelling, it was required to not only increase the power of the devices as the power consumption of the VMs increase. It was also required to define a limit or threshold for the power that describes the maximum allowed power.

$$\frac{\partial PC}{\partial P_V} = f(PC(P_V)) \quad (3)$$

where f is a non-linear function of $PC(P_V)$. This model can be converted to a non-linear or logistic power model by using Taylor expansion series, as follows:

$$\frac{\partial PC}{\partial P_V} = f(PC(P_V)) = K_0 + k_1 PC(P_V) + k_2 PC(P_V)^2 + \dots \quad (4)$$

Retaining terms at second degree. There are some assumptions to be made to settle the final expression.

Theorem 1. *There is no change in the PC at the initial state, that is, $\frac{\partial PC}{\partial N} = 0$ when $PC(P_V) = 0$, substitute in Equation (4), yields $k_0 = 0$.*

Theorem 2. *There is no change in the PC when the model reaches its maximum power value PC_m . Hence, $\frac{\partial PC}{\partial P_V} = 0$ when $PC(P_V) = PC_m$, substitute in Equation (4) to produce:*

$$k_1 PC_m + k_2 PC_m^2 = 0 \quad (5)$$

which produces:

$$k_2 = \frac{-k_1}{PC_m} \quad (6)$$

Returning Equation (12) in Equation (4) produces:

$$PC(P_V) = \frac{PC_m}{1 + \left(\frac{PC_m - PC_o}{PC_o}\right) e^{-K P_V}} \quad (7)$$

Proof of Theorem 1: As per the logistic model, there are two points where the change in the behaviour makes no growth waiting for the network load to increase via operating the VMs. Hence logically, the start of the model that assumed to be the server PC in the idle mode of operation shows no power increment. At this point, there is no change in PC at the initial state, that is, $\frac{\partial PC}{\partial P_V} = 0$ when $PC(P_V) = 0$.

Substituting into Equation (4), produces:

$$K_o + k_1 \cdot 0 + k_2 \cdot 0^2 = 0 \quad (8)$$

By simplifying, it can be observed that $K_o = 0$.

Proof of Theorem 2: As the PC of the server cannot exceed a maximum value, which is specified for each server as per the data sheet of each device. It is inevitable that no change in PC occurs when the model reaches its maximum power value PC_m . Mathematically, this can be expressed as $\frac{\partial PC}{\partial P_V} = 0$ when $PC(P_V) = PC_m$.

Substituting this condition into Equation (4):

$$k_1 PC_m + k_2 PC_m^2 = 0 \quad (9)$$

Divide each side by PC_m :

$$k_1 + k_2 PC_m = 0 \quad (10)$$

or

$$k_1 = -k_2 PC_m \quad (11)$$

Solve for k_2

$$k_2 = \frac{-k_1}{PC_m} \quad (12)$$

Case 3. The power consumption of the VMs is linearly or exponentially related to the number of processed events. In the case of mobile network, the understood parameter is the resource blocks (RB) or number of bits [26, 27].

Note that partial differential equations (PDEs) and ordinary differential equations (ODEs) are the key

mathematical methods that play a fundamental role in modelling the real world problems, which are related to multi-aspect physics in various fields of science. These equations are grounds for which truth looks that systems of the like nature are prone to becoming powerful and behaving randomly considering that they are responded by many independent variables (for the case of PDEs) or a single one of independent variables (in the case of ODEs one). PDEs and ODEs is the tool that researchers use to mathematically formulate and solve problem about complex issues like diffusion of heat, wave motion, and growth dynamics. That's a way they could make predictions and operations easier, just based on data processing. In other words, such standards are found to really the basis upon which the empirical facts or theoretical concepts can be placed together as a result, they are essential structure in scientific research and engineering systems appropriate.

3.2 | Resource blocks considerations

The modulation type, represented as mod, determines the number of bits L_{mod} in each resource element. For instance, in the case of QPSK modulation, the number of bits is $L_{qpsk} = 2$. Moreover, the value of L_{qam} in 16-QAM is 4. The number of bits in each resource block is determined by multiplying the sub-carriers' number in the RB (N_{sc}^{RB}) by the number of symbols in the RB (N_{sym}^{RB}).

$$L_{RB}^{mod} = L_{mod} N_{sc}^{RB} N_{sym}^{RB} \quad (13)$$

In order to get the total number of bits of the total RBs, one must multiply the modulation order of each RB (L_{RB}^{mod}) by the total number of RBs (RBs).

$$L_{RBs}^{mod} = L_{RB}^{mod} RBs \quad (14)$$

Hence,

$$RBs = \frac{L_{RBs}^{mod}}{L_{RB}^{mod}} \quad (15)$$

The current regulations allow for the modelling of virtual machine power consumption in relation to the quantity of RBs or the quantity of bits. The modelling of P_V can be approached similarly to the logistic growth model, as it involves the ability of RBs to maximise the PC of the device to a maximum allowed level, where all the VMs cooperate to reach the maximum consumption of the device, as follows:

$$P_V(RBs) = \frac{P_{V,m}}{1 + \left(\frac{P_{V,m} - P_{V,o}}{P_{V,o}}\right) e^{-\beta RBs}} \quad (16)$$

Note that this formula holds true for one VM, if there is more than VM, the expression ($P_V(RBs)$) is then multiplied by VMs' number within the server.

$$P_V^T(RBs) = P_V(RBs)N \quad (17)$$

where N denotes the total number of server's VMs.

Now it is possible to find the final delay formula in terms of RBs :

$$\tau(PC) = \tau_o e^{\alpha PC(P_V^T(RBs))} \quad (18)$$

4 | DELAY MINIMISATION

In the context of a single server, the objective is to decrease latency by taking into account the number of processed RBs, as well as the maximum and lowest threshold times. In this study, we consider a scenario involving two servers and aim to limit the delay experienced by both servers while adhering to a set of requirements.

$$\begin{aligned} \min \quad & \tau(PC) \\ \text{s.t.} \quad & \\ & RBs_1 + RBs_2 \leq RBs^{th} \\ & PC_{m,1} + PC_{m,2} \leq PC_m^{th} \\ & PC_{o,1} + PC_{o,2} \geq PC_o^{th} \\ & N_1 - N_2 > 0 \\ & N_s, RB_s^v, PC_{m,s}, PC_{o,s}, N_1, N_2 \geq 0 \end{aligned} \quad (19)$$

The first constraint demonstrates that the total RBs allocated to the two servers $s = 1$ and $s = 2$ does not beyond a predetermined threshold value denoted as RBs^{th} as each server has a specific capacity for processing RBs, which is determined by its manufacturing features. The second restriction specifies the maximum power consumption for the both servers doesn't exceed a threshold value PC_m^{th} . The third restriction provides information about the total amount of power regarding the initial consumption. In the fourth constraint, we have that $N_1 > N_2$, this concept operates under the assumption that server 1 experiences a higher load or possesses the capacity to process a bigger number of RBs, resulting in the installation of a greater number of VMs. Subsequently, the ultimate set of equations representing the partial derivatives, in conjunction with the Lagrangian, can be expressed as follows:

$$\begin{aligned} \frac{\partial L}{\partial RBs_1} &= \alpha \tau_{o,1} \\ & \frac{PC_{m,1}}{1 + \left(\frac{PC_{m,1} - PC_{o,1}}{PC_{o,1}} \right) e^{-K \left(\frac{P_{V,m,1}}{1 + \left(\frac{P_{V,m,1} - P_{V,o,1}}{P_{V,o,1}} \right) e^{-\beta RBs_1}} \right)}} \\ -\lambda_1 &= 0 \\ \frac{\partial L}{\partial RBs_2} &= \alpha \tau_{o,2} \\ & \frac{PC_{m,2}}{1 + \left(\frac{PC_{m,2} - PC_{o,2}}{PC_{o,2}} \right) e^{-K \left(\frac{P_{V,m,2}}{1 + \left(\frac{P_{V,m,2} - P_{V,o,2}}{P_{V,o,2}} \right) e^{-\beta RBs_2}} \right)}} \\ -\lambda_1 &= 0 \\ \frac{\partial L}{\partial PC_{m,1}} &= \alpha \tau_{o,1} \\ & \frac{1}{1 + \left(\frac{PC_{m,1} - PC_{o,1}}{PC_{o,1}} \right) e^{-K \left(\frac{P_{V,m,1}}{1 + \left(\frac{P_{V,m,1} - P_{V,o,1}}{P_{V,o,1}} \right) e^{-\beta RBs_1}} \right)}} \\ -\lambda_2 &= 0 \\ \frac{\partial L}{\partial PC_{m,2}} &= \alpha \tau_{o,2} \\ & \frac{1}{1 + \left(\frac{PC_{m,2} - PC_{o,2}}{PC_{o,2}} \right) e^{-K \left(\frac{P_{V,m,2}}{1 + \left(\frac{P_{V,m,2} - P_{V,o,2}}{P_{V,o,2}} \right) e^{-\beta RBs_2}} \right)}} \\ -\lambda_2 &= 0 \\ \frac{\partial L}{\partial PC_{o,1}} &= \alpha \tau_{o,1} \\ & \frac{PC_{m,1} - PC_{o,1}}{PC_{o,1} \left(1 + \left(\frac{PC_{m,1} - PC_{o,1}}{PC_{o,1}} \right) e^{-K \left(\frac{P_{V,m,1}}{1 + \left(\frac{P_{V,m,1} - P_{V,o,1}}{P_{V,o,1}} \right) e^{-\beta RBs_1}} \right)} \right)} \\ -\lambda_3 &= 0 \end{aligned}$$

$$\frac{\partial L}{\partial PC_{o,2}} = \alpha \tau_{o,2}$$

$$PC_{m,2} - PC_{o,2}$$

$$PC_{o,2} \left(1 + \left(\frac{PC_{m,2} - PC_{o,2}}{PC_{o,2}} \right) e^{-K \left(\frac{P_{V,m,2}}{1 + \left(\frac{P_{V,m,2} - P_{V,o,2}}{P_{V,o,2}} \right) e^{-\beta RB_2}} \right)} \right)$$

$$-\lambda_3 = 0$$

By solving the system of equations in conjunction with the initial constraints, the optimal values of the variables τ_1 , τ_2 , RBs_1 , RBs_2 , $PC_{max,1}$, $PC_{max,2}$, $PC_{o,1}$, $PC_{o,2}$, N_1 , N_2 may be determined. These values will minimise the objective function ($\tau_1 + \tau_2$) while ensuring compliance with the specified constraints. As can be seen, the set of equations are not trivial to solve mathematically to find the optimisation parameters' values. Hence, numerical methods or software tools such as mathematical optimisation solvers are being commonly employed to obtain the solutions.

5 | PROBLEM ALGORITHMS

Considering the computer science/engineering domains, it is vital to realistically adapt to the evolution of mathematical models into algorithms. These algorithms are employed to simulate the delay in network systems while considering different constraints and parameters, thus enhancing the system performance. This work proposed two algorithms for delay modelling and optimisation, as follows:

A. Algorithm for Delay Modelling, in Figure 2 is summarised as follows:

1. Define inputs: RBs , PC_m , PC_o , $P_{V,m}$, $P_{V,o}$, N , α , β , K .
2. Calculate power consumption $PC(P_V)$ using Equation (17).
3. Compute delay $\tau(PC)$ using the power consumption obtained in the previous step.
4. Output the delay τ .

The delay model is essentially a power model that reveals how delay relates to measured and actual data of delay. It posits an exponential relation between the workload, server delay, server energy, and the number of VMs. The model displays exponential rise in workload and power consumption as the dynamic workload increases on different networks and servers. The model is mathematically defined as a growth model, wherein VM power consumption is formulated using a logistic equation.

B. Algorithm for Delay Minimisation, in Figure 3 is summarised as follows:

1. Define inputs: RBs^{th} , PC_m^{th} , PC_o^{th} , α , β , K .

2. Define initial values for RBs_1 , RBs_2 , $PC_{max,1}$, $PC_{max,2}$, $PC_{o,1}$, $PC_{o,2}$, N_1 , N_2 .
3. Use optimisation solver to minimise the delay function $\tau_1 + \tau_2$ subject to the given constraints.
4. Output the optimal values for RBs_1 , RBs_2 , $PC_{max,1}$, $PC_{max,2}$, $PC_{o,1}$, $PC_{o,2}$, N_1 , N_2 .

The presented Algorithm 3 non-linear and non-convex objective function of server delay and constraint as a combination of different network variables with complete mathematical solutions using Active-set, Interior Point, and Sequential Quadratic Programming (SQP). That is highly dependent on the algorithm selected and parameters, but mostly, numerical solutions are used. The delay model is highly associated with PC, the number of VMs, scalability and real time application, which exhibit high complexity behaviour in the computation aspect. The delayed-task work overload is one of the complicating factors in the optimisation problem since their work delay growth is exponential to the workload. Thus, cloud computing and high-performance power demand real-time performance.

6 | RESULTS AND DISCUSSION

6.1 | Delay performance

The proposed model has demonstrated significant alignment with the practical approaches discussed in the works of [21, 26]. Figure 4 illustrates the proposed delay model in relation to various values of α , ranging from 0.04 to 0.08.

However, the values of α in Figure 5 have been modified to fall within the range of 0.01–0.05. The modification has resulted in an increased alignment with the findings of [21] and the study conducted by the authors of [26], specifically above the top half region of the figure. However, the proposed model has been subject to various constraints imposed by modifying parameters, ranging from fine-tuning of the constant parameters, through the number of RBs, to the larger values such as the initial latency of the servers. Modifying the value of each dependent parameter results in a distinct behaviour that differs from the one now observed. The aforementioned versatility provides a dependable framework that is applicable to many scenarios, states, server types, and applications Table 1.

For example, adjusting the value of RBs can impact the level of matching between the outcomes of the proposed model and those of previous studies, while keeping all other parameters constant, as depicted in Figure 6.

6.2 | Optimisation performance

Three distinct algorithms, namely SQP, Interior-Point, and Active-Set, were utilised to minimise the objective function. The objective function aimed to minimise the sum of the delays of two servers, τ_1 and τ_2 , respectively, in a constrained non-linear optimisation problem. The task entailed the

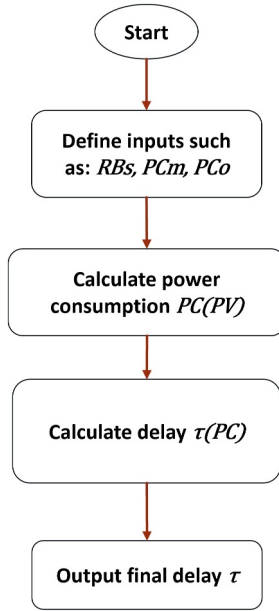


FIGURE 2 Algorithm for delay modelling.

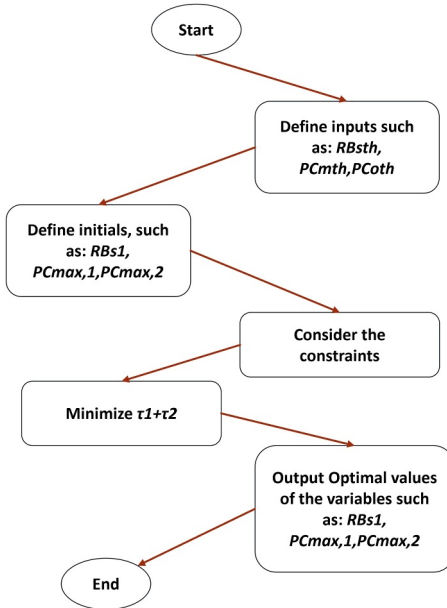


FIGURE 3 Algorithm for delay minimisation.

consideration of eight decision variables, each of which was constrained by both upper and lower bounds. The threshold values and problem-specific constants were predetermined. The findings obtained indicate slight variations in the optimal variable values among the three algorithms. As an example, the SQP approach resulted in minor elevated values for $RBs1$ and $RBs2$, while ensuring that the remaining variables remained within the specified constraints. On the contrary, the utilisation of the Interior-Point algorithm resulted in slight decreased values for $RBs1$ and $RBs2$; nonetheless, it demonstrated compromises in other variables. The Active-Set method yielded a composite of these patterns. The aforementioned results

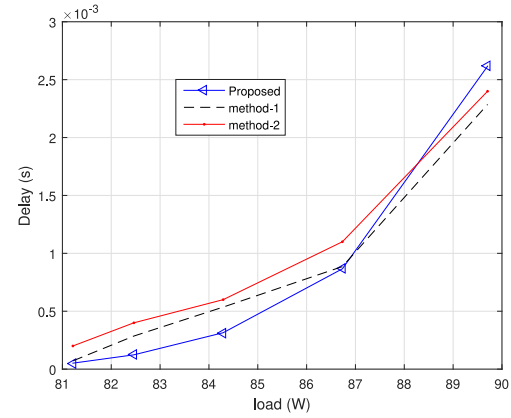
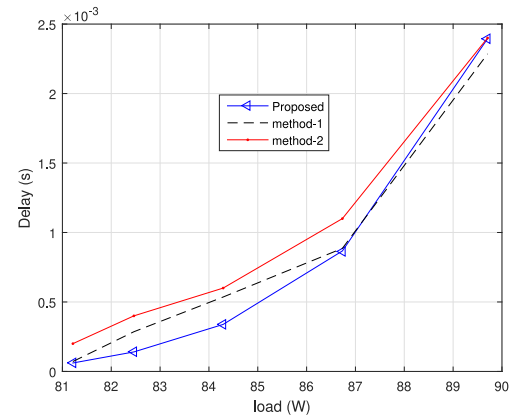
FIGURE 4 Comparison of the delay performance of the proposed method with [21] (method-1) and [26] (method-2); $\alpha = 0.04:0.08$.FIGURE 5 Comparison of the delay performance of the proposed method with [21] (method-1) and [26] (method-2); $\alpha = 0.01:0.05$.

TABLE 1 Model parameters.

Factor	Value	Unit	Factor	Value	Unit
RBs	40	—	$P_{V,m}$	10	W
$P_{V,o}$	1	W	PC_m	130	W
PC_o	80	W	K	0.02:0.08	—
α	0.02:0.06	—	β	0.02:0.06	—
τ_o	2×10^{-6}	s	Height		

emphasise the susceptibility of the optimisation problem to the selection of algorithm and emphasise the significance of meticulously choosing a suitable optimisation approach to effectively fulfil certain objectives and restrictions.

The following parameters has been used in the optimisation problem: $RB_{tb} = 80$; $PC_{max,tb} = 200$; $PC_{o,tb} = 80$; $N1 = 10$; $N2 = 7$; $\alpha = 0.001$; $\tau_{o,1} = 0.01$; $K = 0.001$; $P_{VM,max,1} = 5$; $P_{VM,o,1} = 1$.

Figure 7 shows the influence of different optimisation parameters on the objective function value by utilising three algorithms: SQP, Interior Point, and Active Set as indexed in the x -axis, respectively.

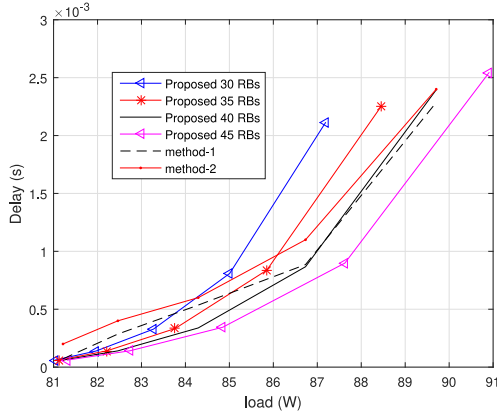


FIGURE 6 Comparison of the delay performance of the proposed method with [21] (method-1) and [26] (method-2) using different values of RBs.

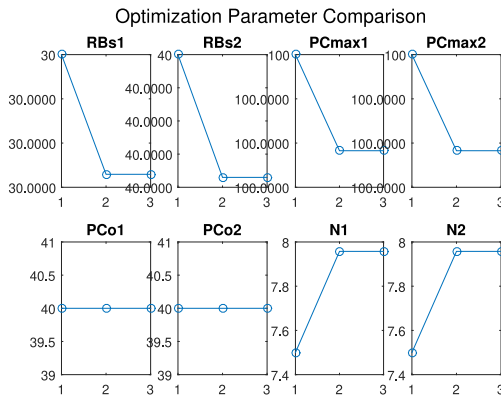


FIGURE 7 Impact of various optimisation settings on the objective function value, employing the SQP, interior point, and active set algorithms.

By customising the constraints to the ones shown in Equation (20), Figure 8 is produced.

$$\begin{aligned}
 A \times RBs_1 + RBs_2 &\leq RBs^{th} \\
 PC_{m,1} + B \times PC_{m,2} &\leq PC_m^{th} \\
 B \times PC_{o,1} + PC_{o,2} &\leq PC_o^{th} \\
 N_1 - 1.5 \times AN_2 &> 0
 \end{aligned} \tag{20}$$

The purpose of customising the constraints is to demonstrate the adaptability of the optimisation problem to various contexts, for example, in 20, $A = 1.5$ and $B = 1.3$. The introduction these scaling factors can offer distinct advantages in specific cases when compared to the non-customised case. First, achieving balance impact through the manipulation of scaling factors, it is possible to exert control over the relative contributions of various variables to the objective function, hence influencing the overall balance of impact. Within the given limitations, the scaling variables exert an influence on the comparative significance of RBs , PC_{max} , PC_o , $N1$ and $N2$. This feature enables the user to assign higher priority to a

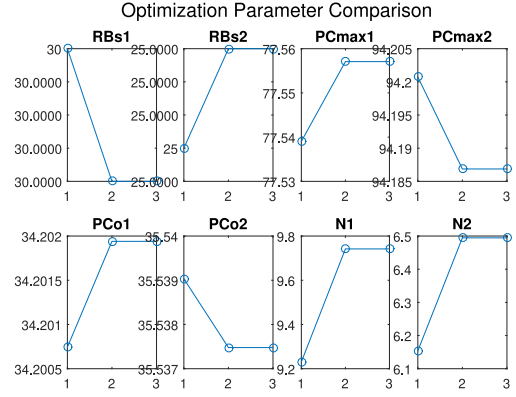


FIGURE 8 Impact of various optimisation settings on the objective function value, employing the SQP, interior point, and active set algorithms using the constrains mentioned in (20).

certain group of variables over another, taking into account the specific requirements of their optimisation problem. Second, achieving fine tuning, depending on the specific application, this shall possess a more comprehensive understanding of the appropriate treatment of various variables. The utilisation of customisation can facilitate the alignment of the optimisation process with domain-specific knowledge and the requirements of the business. Third, the utilisation of scaling factors can also offer a mean to loosen or tighten limitations. For example, the act of increasing the scaling factor associated with a constraint results in a heightened level of stringency, whereas reducing the scaling factor leads to a relaxation of the constraint. And finally, customisation enables the adaptation of the optimisation problem to various scenarios. In instances when the importance of specific variables, following vendors, operators and ORAN policy regulations, exhibits variability across different settings, it is possible to adapt the scaling factors accordingly to enhance the versatility of the optimisation technique.

Nevertheless, it is imperative to bring caution when employing scaling variables, as they have the ability to inject subjectivity and inadvertently produce biased or undesired outcomes. The selection of scaling factors must be grounded in a comprehensive understanding of the problem and their influence on the outcomes of optimisation should be meticulously assessed and confirmed. Furthermore, it is crucial to record the underlying justification for the modification in order to provide transparency and replicability in the optimisation procedure.

In order to ascertain the convexity, it is necessary to evaluate the convexity of both the objective function and the constraints. The objective function is a non-linear function that incorporates exponential terms and divisions. The presence of exponential terms in the objective function results in non-convexity, as the exponential function itself is intrinsically non-convex. Subsequently, the constraints pertain to the formation of linear combinations of variables. Each constraint can be represented as a linear equation, which is a form that exhibits convexity. Nevertheless, the inclusion of a

non-convex objective function results in the problem being classified as non-convex in its entirety. In the context of non-convex optimisation problems, the task of identifying the global minimum might provide difficulties due to the presence of numerous local minima. Subsequently, algorithms designed for non-convex problems may not guarantee discovering the global optimum, unless the initial solution is carefully selected, which can exert a substantial influence on the outcomes.

7 | CONCLUSIONS AND FUTURE TRENDS

In summary, this research examines the intricate nature of a resource allocation problem involving multiple variables in network environments, with a specific focus on addressing concerns related to QoS in ORAN. The main objective is to reduce the overall latency among several servers while also considering network limitations. The delay of each server is described by an exponential non-linear function, which results in inherent non-convexity in the objective function. In the meantime, the limitations encompass linear combinations that incorporate resource block assignments, power consumption, and virtual machine counts. These constraints guarantee that resource allocation adheres to practical boundaries and upholds equitable distribution among servers. Nevertheless, the inclusion of a non-convex objective function significantly increases the complexity of the optimisation problem, requiring the utilisation of specialised techniques and methodologies to identify potential solutions. Therefore, the utilisation of the Lagrange multiplier technique provides a mathematical solution to this particular difficulty. The problem is addressed by utilising and numerically comparing three algorithms: Active-set, Interior Point, and SQP. It is worth noting that the objective function of total delay is intricately connected to the overall power consumption of servers. Prior to the optimisation of the overall delay, a delay model is provided and compared to two previous research studies that utilise experimental and real-time data. The suggested model exhibits comparability with existing studies and offers more flexibility and integration with diverse server characteristics and network factors utilised in various research endeavours.

In the Future, it is preferred to expand the area of comparison by adding space complexity, convergence speed, robustness, sensitivity analysis, scalability, implementability, and numerical stability to time complexity. Space complexity is another aspect of algorithm analysis that ensures that the algorithm has enough memory to deal with large-scale problems; and convergence rate is the measure of the rate at which an algorithm is able to generate an optimal solution. Stability is necessary for dealing with noisy, or limited, data, model parameter errors, and model dynamics variations. Sensitivity analysis refers to the judgement of the solution's fragility when the input parameters or constraints change while scalability deals with the consideration of the algorithm's capability of

execution for growing problem sizes. The efficiency category reflects the difficulty in a real-life implementation of the algorithm and numerical stability is greatly important for solving complicated problems.

AUTHOR CONTRIBUTIONS

This research paper is a collaborative effort of multiple authors, each contributing significantly to various aspects of the study, as below: Basit N. Khalaf: Contributed to the conceptualisation and design of the study. Played a key role in developing the mathematical models. Wisam Hasan Ali: Involved in data collection and analysis. Raad S. Alhumaima: Assisted in the formulation and optimisation of the non-convex objective functions. Haider Ali Jasim Alshamary: Review of related works and literature.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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